

EE115 Linear Circuit Analysis (All sections)
Mid Term – II (FALL 2015)

Maximum Marks: 45

Instructor: Dr. Saima Zafar & Ms. Beenish Fatima

Total Time: 1 hr

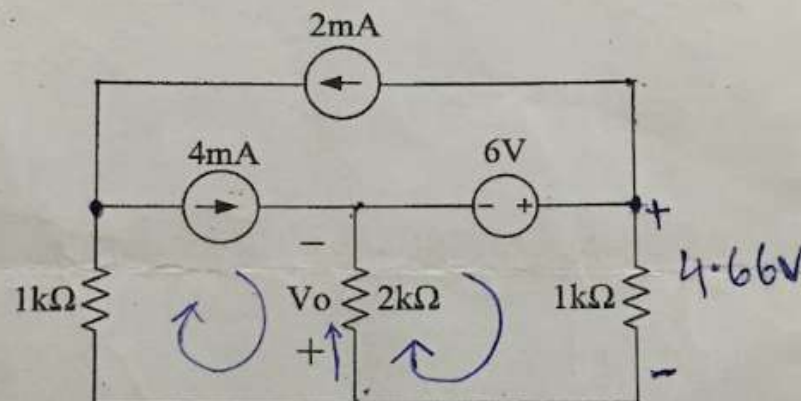
Dated: 7th November, 2015

INSTRUCTIONS: Carefully read the paper and attempt each question (only once) by using the technique mentioned in it. Show all calculations, direct answers are not acceptable. Make assumptions if you think some data is missing, but clearly mention your assumption. Your answers should be approximated up to two decimal places. Attempt all parts of a question together.

Question # 1 [15 marks]

Class Learning Objective: An ability to apply mesh current method to find the current/ voltage in a resistive circuit.

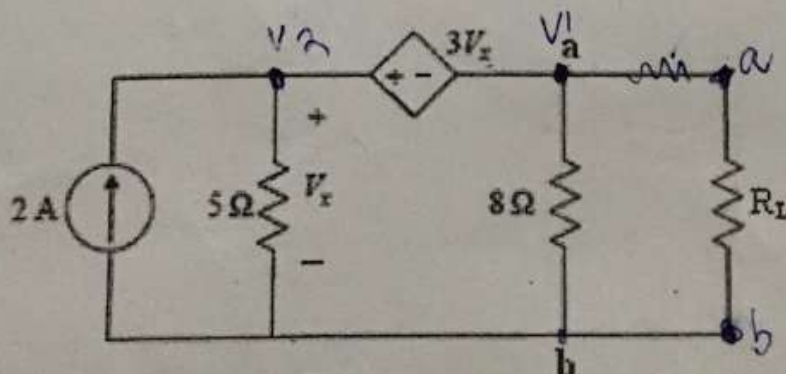
- Find V_0 in the circuit shown below using **mesh analysis**.
- Use the result of part (a) to find the power associated with the 4mA current source.



Question # 2 [15 marks]

Class Learning Objective: An ability to determine the Thevenin equivalent circuit and compute maximum power delivered to a load resistor.

In the circuit shown below, find the value of R_L that will absorb maximum power from the circuit. Find also the value of this maximum power.



Question # 3 [15 marks]

Class Learning Objective: An ability to use the principle of superposition to find current/ voltage in a resistive circuit containing more than one independent source.

Use **superposition** to find the voltage V in the circuit given below:

